1. Clinical Question & Core Hypothesis

Clinical Question

The ontologist's single-path analysis highlights that PROS1 is expressed in tumors and is associated with Ki-67, raising the question of how PROS1-related biology mechanistically links to tumor proliferative state and clinical progression. 2) Tier 1 path used: the single provided path (Tumors → PROS1 → Ki-67). I built on this because it directly connects a tumor-expressed coagulation-related factor (PROS1) to proliferation (MKI67), forming a tight mechanistic backbone. 3) Tier 2 evidence: none was provided; supportive context comes only from the path's implication that imaging (ultrasound) does not perturb gene expression, so observed expression–imaging associations likely reflect biology rather than imaging artifacts. 4) Endpoint choice: I selected progression-free interval (PFI) because the question concerns whether PROS1-driven proliferative programs in a particular spatial microenvironment accelerate disease progression (PFI captures time to progression/recurrence). 5) Deliberate exclusions: The ontologist mentioned ultrasound, but the project's available imaging features are limited to TIL density, TLS-like aggregates, and stromal-to-tumor ratio (TSR); I therefore chose TSR as the most biologically relevant imaging correlate of PROS1's stromal/coagulation role and excluded ultrasound from the core hypothesis to conform to the dataset's imaging feature set. No additional genes were invented or included in the core hypothesis to keep the testable mechanistic focus?

Working Hypothesis

In bladder tumors with high stromal-to-tumor ratio (TSR), elevated tumor cell PROS1 expression promotes paracrine signaling within the stromal compartment via TAM receptor–mediated pathways, driving tumor cell proliferation as indicated by increased MKI67 expression. If this TSR-dependent PROS1–Ki-67 axis operates, TSR-high/PROS1-high tumors will have shorter progression-free interval (PFI) compared with TSR-high/PROS1-low or TSR-low groups.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	PROS1, MKI67
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

PROS1: PROS1 is expressed in tumors and is associated with Ki-67, raising the question of how PROS1-related biology mechanistically links to tumor proliferative state and clinical progression.

MKI67: MKI67), forming a tight mechanistic backbone.

3. Clinically Testable Predictions

Prediction 1: retrospective analyses of bladder cancer cohorts with measured TSR, PROS1, MKI67, and PFI;

Prediction 2: spatially resolved profiling to map PROS1 and MKI67 within TSR-defined regions;

Prediction 3: in vitro co-culture systems (bladder cancer cells with fibroblasts) manipulating PROS1 levels and TAM signaling to assess MKI67 induction;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions